

Haoxuan Sun

PhD Candidate in Physics · Experimental Particle Physics & Machine Learning

haoxuansun0616@gmail.com | +86 158 9568 6963 | Hong Kong SAR, China
[Google Scholar](#) | [INSPIRE-HEP](#) | [GitHub](#) | [GitLab \(CERN\)](#) | [franksun0616.github.io](#)



EDUCATION

Hong Kong University of Science and Technology
Doctor of Philosophy in Physics
Hong Kong SAR, China · Expected May 2028

Brown University
Master of Science in Physics
Providence, RI, USA · May 2025 · GPA 3.83 / 4.00

Macalester College
Bachelor of Arts in Physics and Mathematics
St. Paul, MN, USA · May 2023 · GPA 3.92 / 4.00

HONORS & AWARDS

Doctoral
Hong Kong PhD Fellowship Scheme (HKPFS)
Redbird Scholarship

Undergraduate
Physics Honors Program
Turck Presidential Scholarship (four times)
Dean's List (all semesters)

TECHNICAL SKILLS

PROGRAMMING
Python, Java, JavaScript, R

SCIENTIFIC COMPUTING
Athena, MadGraph, Pythia, Salt, TensorFlow, Mathematica, Jupyter Notebook, Google Colab

ENVIRONMENTS & TOOLS
Linux, Microsoft Windows, LaTeX, Microsoft Office

LANGUAGES

Mandarin (native) · English (fluent) · Spanish (fluent)

PROFILE

Physics PhD candidate working at the intersection of experimental particle physics and machine learning. Member of the ATLAS Collaboration, contributing to Higgs boson measurements in the $H \rightarrow ZZ \rightarrow 4\ell$ channel and to deep-learning electron identification for Run-3 data.

Research interests: Higgs boson properties and EFT interpretations · machine learning for particle reconstruction and calibration · cosmology and nucleosynthesis.

CURRENT RESEARCH

Decay STXS Implementation in the $H \rightarrow ZZ \rightarrow 4\ell$ Sector Sep 2025 – Present
ATLAS Collaboration (RML Subgroup) · HKUST

- Implementing fiducial cut definitions for Higgs boson decay STXS inside the ATLAS Athena framework, extending the derivation and ntuple workflow for the $H \rightarrow ZZ \rightarrow 4\ell$ analysis chain.
- Investigating SMEFT Higgs coupling modifications through large-scale Monte Carlo generation with MadGraph and Pythia, producing cross-section and yield distributions as a function of EFT operator coefficients.

DNN-Based Electron Identification for Run-3 ATLAS Data Jan 2026 – Present
ATLAS Collaboration (EGamma Subgroup) · HKUST

- Training a DNN-based electron identification menu on Run-3 data to replace the current Bayes classifier, focusing on low- p_T performance relevant to $H \rightarrow 4\ell$ signatures.
- Preparing training samples via the Tag-and-Probe framework and migrating the trained model from TensorFlow to the Salt framework for improved flexibility and ATLAS software integration.

PREVIOUS RESEARCH

Neural Network Application for Jet Calibration Jan 2024 – Nov 2025
Advisor: Dr. Jennifer Roloff · Brown University

- Developed neural-network models for jet calibration, integrating a Particle Flow Network (PFN) to reconstruct jet constituents.
- Performed hyperparameter tuning and validation to identify optimal configurations; implemented a dual-stream framework and knowledge-distillation scheme that improved calibration accuracy.
- Integrated the models into the existing calibration framework and validated results against CMS open data, demonstrating consistent performance with minimal residual error.
- Released the code for public use with configurable settings for multiple input jet data types.

PUBLICATIONS & PREPRINTS

1.

West, C., Heger, A., Côté, B., Serxner, L., & **Sun, H.** (2025). The Impact of Initial Composition on Massive Star Evolution and Nucleosynthesis. *The Astrophysical Journal*, 987(1), 51.

2.

Ansari, R., Marques, G. A., Marriner, J. P., Perdereau, O., Pinetti, E., Robinthal, L., Stebbins, A., **Sun, H.**, Timbie, P., Tucker, G. S., Doyle, E., Chu, J., Karaalp, E. R., Chen, X., Li, J., & Wu, F. (2025). The Tianlai–WIYN North Celestial Cap Redshift Survey. [arXiv:2512.23899](#) [[astro-ph.CO](#)].

3.

Hu, Z., Gao, Z., **Sun, H.**, Chen, C., & Yang, X. (2024). Renormalizable Cosmology Based on Gauss–Bonnet Theory with Torsion. [arXiv:2403.07250](#).

4.

Sun, H. (2023). Machine Learning Refinements to Metallicity-Dependent Isotopic Abundances. *Macalester Journal of Physics and Astronomy*, 11(1), [Article 14](#).

SELECTED TALKS & POSTERS

AoE Mini-Symposium 2026 <i>Hong Kong SAR, China</i>	Presentation · Mar 2026
Brown University Department of Physics Annual Poster Session <i>Providence, RI, USA</i>	Poster · Nov 2024
APS April Meeting 2024 <i>Sacramento, CA, USA</i>	Poster · Apr 2024
APS March Meeting 2024 <i>Minneapolis, MN, USA</i>	Poster · Mar 2024
Solar Physics Division Annual Meeting (AAS) <i>Minneapolis, MN, USA</i>	Poster · Aug 2023
Macalester Showcase of Summer Creative Endeavors, Research and Scholarship <i>St. Paul, MN, USA</i>	Poster · Sep 2022

TEACHING EXPERIENCE

Teaching Assistant		Preceptor	
<i>Hong Kong University of Science and Technology</i>		<i>Macalester College, St. Paul, MN</i>	
PHYS 4191 — Physics Capstone Project	Fall 2025	MATH 471 — Topology	Spring 2023
Grader		PHYS 127 — Introductory Physics II	Spring 2022
		PHYS 126 — Introductory Physics I	Spring 2022
		PHYS 227 — Principles of Physics II	Fall 2021
		PHYS 226 — Principles of Physics I	Fall 2021
<i>Brown University, Providence, RI</i>			
PHYS 1420 — Advanced Quantum Mechanics	Spring 2025		
PHYS 0050 — Foundations of Mechanics	Fall 2024		
Grader			
<i>Macalester College, St. Paul, MN</i>			
PHYS 227 — Principles of Physics II	Fall 2022		